Tracing Circuits in Qwen3: A Faithful Reimplementation and Reproduction of Attribution Graphs*

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July 12, 2026

Abstract

Attribution graphs decompose a language model's forward pass on a single prompt into causal, weighted interactions between sparse dictionary features, and have been used to map the "biology" of a frontier model. We reimplement the full methodology from scratch for the Qwen3 model family — local replacement models, batched edge attribution, influence-based pruning, and constrained-patching interventions — using the open-source circuit-tracer library only to load pretrained per-layer transcoders. The reimplementation is verified numerically equivalent to circuit-tracer: identical node sets, matching edge weights, identical pruned graphs, and matching constrained-steering effects on shared inputs. We then reproduce the multilingual-antonyms case study of Lindsey et al. [3] on Qwen3-4B — shared cross-lingual circuitry and the paper's operation, operand, and language swap interventions — following its protocols, and add a same-recipe cross-scale comparison (4B vs. 8B). The circuit structure reproduces; the operand swap reproduces at 1.5–12× the paper's steering strengths; the language swap reproduces in two of three directions at the paper's own scale; and the operation swap does not reproduce under the protocol's frozen attention patterns — a probe that lets patterns re-form recovers the paper's exact outcome tokens, identifying the channel as QK-mediated, the same interaction the paper flags as invisible to its method. The cross-scale overlap-growth claim does not survive a register-diverse evaluation corpus.

1 Introduction

Mechanistic interpretability aims to reverse-engineer the computations of neural networks into human-understandable algorithms [4, 5]. For large language models a central obstacle is that the natural unit of computation is not the neuron: models appear to represent many more concepts than they have neurons, storing them in *superposition*, so individual neurons are polysemantic and resist interpretation [6]. Explaining a behaviour mechanistically therefore requires both a better feature basis and a way to trace, on concrete prompts, how those features causally interact.

Dictionary learning addresses the first requirement. Sparse autoencoders (SAEs) decompose activations into an overcomplete basis of sparsely-firing, substantially more monosemantic features [7, 8], scale to frontier models [9], and benefit from discontinuous activation functions such as JumpReLU [10]. An SAE, however, only re-describes the representation at one point in the network: it says what is encoded, not how it is computed. *Transcoders* [11] instead train the same sparse dictionary architecture to imitate an MLP's input–output map, so that each feature has both an input action (its encoder row) and an output action (its decoder column) and the MLP's computation factors through an interpretable sparse bottleneck. Cross-layer variants let one feature write to all subsequent layers [1].

*Code: `llm-circuits` repository. Reference method: Ameisen et al. [1]; reference implementation: Hanna et al. [2].

The second requirement — tracing interactions — has traditionally been met with causal patching. Manual activation patching over attention heads and MLPs isolated, e.g., the indirect-object-identification circuit in GPT-2 [12]; ACDC automated the search [13]; attribution patching replaced interventions with cheap gradient approximations [14]; and sparse feature circuits lifted patching from model components to SAE features [15]. *Attribution graphs* [1] sharpen this line of work into a per-prompt causal graph over transcoder features. Replacing MLPs with transcoders, freezing attention patterns and normalisation denominators at their on-prompt values, and inserting error nodes yields a *local replacement model* that reproduces the underlying model's forward pass exactly while being linear in the feature activations; *direct* effects between features are then exact linear coefficients obtained by backpropagation rather than approximations, and the resulting dense graph is pruned to an interpretable subgraph whose hypotheses are validated by steering interventions on the real model. The companion paper applies the method across dozens of behaviours of Claude 3.5 Haiku [3], and the circuit-tracer library [2] open-sources the pipeline for open-weight models. Together these form the reference for the present work.

This report makes two contributions. **First**, we reimplement the entire pipeline from scratch for HuggingFace Qwen3 models [16]: capture of frozen constants, the local replacement forward pass, batched edge attribution via `autograd`, influence-based graph pruning, an interactive graph explorer, and constrained-patching feature interventions. circuit-tracer is used *only* to load pretrained per-layer transcoders [17]; no attribution, pruning, or intervention machinery is imported. A dedicated verification suite establishes numerical equivalence with circuit-tracer on shared inputs — identical node sets, edge weights matching to floating-point noise, identical influence rankings and pruned graphs, and matching constrained-steering deltas (Appendix A). **Second**, we reproduce the *multilingual antonyms* case study of Lindsey et al. [3] on Qwen3-4B: a shared cross-lingual "say-large" core with language-specific input and output stages, the paper's operation/operand/language swap interventions under its constrained-patching protocol, and cross-lingual feature overlap as a function of scale, compared against Qwen3-8B with same-recipe transcoders. Where Qwen3 forces protocol adaptations — chat templates for an instruct model, steering strengths at which a 4B model responds only weakly — we document the adaptation explicitly and separate model differences from method failures.

Section 2 presents the method; Section 3 reports the verification and the reproduction; Section 4 discusses what transfers across models and what does not.

2 Method

Throughout, the underlying model is a pre-norm decoder-only transformer with L layers. We write $\boldsymbol{x}_c^\ell \in \mathbb{R}^d$ for the residual stream at token position c before layer ℓ's MLP, $\tilde{\boldsymbol{x}}_c^\ell$ for the normalised MLP input,[1] and $\boldsymbol{y}_c^\ell = \mathrm{MLP}^\ell(\tilde{\boldsymbol{x}}_c^\ell)$ for the MLP output added back into the stream. Logits are $\mathrm{logit}_t = \boldsymbol{u}_t^\top \mathrm{norm}(\boldsymbol{x}^{L+1})$ with unembedding rows $\boldsymbol{u}_t$.

2.1 Transcoders: sparse dictionaries for MLP computation

An SAE learns an overcomplete sparse code $\boldsymbol{f}(\boldsymbol{x}) = \sigma(W_{\mathrm{enc}}\boldsymbol{x} + \boldsymbol{b}_{\mathrm{enc}})$ with linear decoder $\hat{\boldsymbol{x}} = W_{\mathrm{dec}}\boldsymbol{f} + \boldsymbol{b}_{\mathrm{dec}}$, trained to reconstruct its own input under a sparsity penalty. A *transcoder* keeps this architecture but changes the target: it is trained to predict the MLP's *output* from the MLP's *input*, so that the learned features mediate the layer's computation rather than merely re-encode its input. In the per-layer form (PLT) used in this work, layer ℓ has feature activations and reconstruction

$$\boldsymbol{a}_c^\ell = \mathrm{JumpReLU}(W_{\mathrm{enc}}^\ell \tilde{\boldsymbol{x}}_c^\ell + \boldsymbol{b}_{\mathrm{enc}}^\ell), \qquad \hat{\boldsymbol{y}}_c^\ell = {W_{\mathrm{dec}}^\ell}^\top \boldsymbol{a}_c^\ell + \boldsymbol{b}_{\mathrm{dec}}^\ell, \qquad \mathrm{JumpReLU}(z)_i = z_i \mathbb{1}[z_i > \theta_i], \tag{1}$$

[1] In Qwen3, RMSNorm: $\tilde{\boldsymbol{x}} = \boldsymbol{x} \oslash \sigma(\boldsymbol{x}) \odot \boldsymbol{\gamma}$ with $\sigma(\boldsymbol{x}) = \sqrt{\|\boldsymbol{x}\|^2/d + \varepsilon}$.

i.e. a linear map on either side of a hard sparse gate. The reference method's default is the *cross-layer* transcoder (CLT), in which a feature encoded at layer ℓ' writes to the MLP outputs of *all* later layers through separate decoders, $\hat{\boldsymbol{y}}_c^\ell = \sum_{\ell' \le \ell} W_{\text{dec}}^{\ell' \to \ell \top} \boldsymbol{a}_c^{\ell'} + \boldsymbol{b}_{\text{dec}}^\ell$; the PLT is the special case $W_{\text{dec}}^{\ell' \to \ell} = 0$ for $\ell' \neq \ell$. Training minimises reconstruction error with a tanh sparsity penalty,

$$\mathcal{L} \;=\; \sum_{\ell=1}^{L} \|\hat{\boldsymbol{y}}^\ell - \boldsymbol{y}^\ell\|_2^2 \;+\; \lambda \sum_{\ell,i} \tanh(c\,\|W_{\text{dec},i}^\ell\|\, a_i^\ell), \tag{2}$$

with the JumpReLU thresholds θ trained through a straight-through estimator. The pretrained Qwen3 transcoders used here are per-layer JumpReLU dictionaries with 163,840 features per layer and no skip connection [17].

2.2 Attribution-graph generation

Local replacement model. Substituting every MLP by its transcoder yields the *replacement model*, which only approximates the underlying model. The attribution graph is instead built on the *local replacement model* for a specific prompt p: (i) MLPs are replaced by transcoders; (ii) attention patterns and normalisation denominators are *frozen* at the values they take in the underlying model's forward pass on p, making attention and normalisation linear maps of the residual stream; and (iii) an *error node* is added at every (position, layer),

$$\boldsymbol{e}_c^\ell \;=\; \boldsymbol{y}_c^\ell - \hat{\boldsymbol{y}}_c^\ell, \tag{3}$$

injected as a constant so that every activation and logit of the local replacement model *exactly* equals the underlying model's. By construction the only remaining nonlinearities are the JumpReLU gates on feature pre-activations: between any feature's (constant-scaled) output and any downstream pre-activation, the map is affine.

Nodes. The graph has four node types: one *embedding* node per prompt token; one *feature* node per active feature instance $s = (c_s, \ell_s, i_s)$ with activation $a_s > 0$; one *error* node per (position, layer); and *logit* nodes for the smallest set of candidate output tokens whose cumulative probability reaches 0.95, capped at 10, at the final position. Embedding and error nodes have no incoming edges; logit nodes have no outgoing edges.

Edges. Edge weights are the *direct* linear effects of source on target within the local replacement model. Let $\bar{J}_{(c_s,\ell)\to(c_t,\ell_t)}$ denote the Jacobian of the residual stream at (c_t, ℓ_t) with respect to the stream at (c_s, ℓ) computed with stop-gradients on all MLP/transcoder outputs, attention patterns, and normalisation denominators — i.e. the sum over all paths through the residual stream and the frozen attention OV circuits, but through no other feature's nonlinearity. For a feature source s and a feature target t with encoder row $W_{\text{enc},i_t}^{\ell_t}$, the edge weight is

$$A_{s\to t} \;=\; a_s \sum_{\ell_s \le \ell < \ell_t} \left(W_{\text{dec},i_s}^{\ell_s \to \ell}\right)^\top \bar{J}_{(c_s,\ell)\to(c_t,\ell_t)}\, W_{\text{enc},i_t}^{\ell_t}, \tag{4}$$

which for the PLT collapses to the single $\ell = \ell_s$ term. Embedding and error sources use their output vectors ($\boldsymbol{E}_{\text{tok}}$ and $\boldsymbol{e}_c^\ell$) in place of $a_s W_{\text{dec},i_s}$. For a logit target the input vector is the *vocabulary-demeaned* unembedding direction $\boldsymbol{u}_t - \bar{\boldsymbol{u}}$, applied at the (frozen-)normalised final residual — the demeaned logit is the component that affects the softmax, which is invariant to uniform shifts. Because the local replacement model is affine in the source outputs, incoming edges are *complete*: the pre-activation of any feature node satisfies $h_t = \sum_s A_{s\to t} + \text{const}$, where the constant collects bias-path terms ($\boldsymbol{b}_{\text{enc}}$, $\boldsymbol{b}_{\text{dec}}$) that are not represented as nodes. Attention QK circuits are deliberately outside the graph: nodes influence one another through frozen attention outputs, but effects *on* the patterns themselves are not modelled.

Computation. Edges are computed target-by-target with one backward pass per batch of targets: the target's input vector is injected as the gradient seed at its (layer, position), the backward pass runs through the frozen-linear graph, and every source's edge weight is the inner product of the resulting gradient at the source's write location with the source's output vector, as in Eq. (4). The cost is linear in the number of attributed targets and independent of the number of sources; on dense prompts, targets can be selected adaptively in decreasing order of an incrementally re-estimated influence on the logits, so that the most explanatory feature nodes are attributed first.

2.3 Graph pruning

Raw graphs contain 10^4–10^6 edges, most irrelevant to the output. Pruning keeps the subgraph that carries the bulk of the influence on the logits. Let A be indexed $[t, s]$ (rows = targets) and define the normalised unsigned adjacency

$$\hat{A}_{ts} \;=\; \frac{|A_{ts}|}{\sum_{s'} |A_{ts'}|}, \qquad B \;=\; \sum_{k \geq 1} \hat{A}^k \;=\; (I - \hat{A})^{-1} - I, \tag{5}$$

where the Neumann series converges because the graph is a DAG ($\hat{A}$ is nilpotent); B_{tv} sums the strengths of all directed paths $v \to t$, a path's strength being the product of its normalised edge weights. Each node v receives the probability-weighted logit influence

$$I_v \;=\; \sum_{t \,\in\, \text{logits}} p_t \, B_{tv}, \tag{6}$$

with p_t the model's output probability for logit node t. **Node pruning** keeps the minimal set of nodes, in decreasing I_v, whose cumulative share of $\sum_v I_v$ reaches $\tau_{\text{node}} = 0.8$; embedding and logit nodes are always kept.[2] **Edge pruning** recomputes $\hat{A}'$ and I' on the pruned graph and scores each surviving edge by the influence it feeds into its target,

$$s_{ts} \;=\; \hat{A}'_{ts} \, (I'_t + p_t), \tag{7}$$

($p_t = 0$ for non-logit targets, and $I'_t = 0$ for logit nodes, so the parenthesis is the target's score in either case), again keeping the minimal set with cumulative share $\tau_{\text{edge}} = 0.98$. Finally, internal nodes left dangling — features without both an incoming and an outgoing kept edge, error nodes without an outgoing one — are removed iteratively. Typical graphs shrink by roughly an order of magnitude in nodes and two in edges while retaining ≈80% of the logit influence.

2.4 Interventions

Attribution graphs are hypotheses; they are validated by steering features in the *underlying* model and checking that downstream features and logits move as the graph predicts. Steering feature s multiplicatively by M — $a_s \mapsto M a_s$, so $M{=}0$ ablates and $M{=}-1$ flips the sign[3] — adds the corresponding decoder delta to every MLP output the feature writes to:

$$\Delta \boldsymbol{y}^{\ell}_{c_s} \;=\; (M - 1)\, a_s \, W^{\ell_s \to \ell}_{\text{dec}, i_s}, \qquad \ell_s \leq \ell \leq \ell_e. \tag{8}$$

Under **constrained patching** over a layer range $[\ell_s, \ell_e]$, all MLP outputs within the range are pinned to their clean recorded values before the deltas are added, so the intervention itself is the *only* change to any MLP output in the range (no second-order effects within it, matching the

[2] The paper's prose also protects error nodes; the reference implementation (and ours, verified equivalent to it) lets error nodes be pruned.

[3] Our code parameterises the same operation by the additive multiple $m = M - 1$ of the clean activation ($m{=}-1$ ablate, $m{=}-2$ flip); circuit-tracer's API takes the absolute target value $M a_s$. All strengths in this report are quoted in the paper's multiplicative M convention.

graph's direct-effect semantics); computation resumes normally after ℓ_e. Attention patterns are frozen to their clean values throughout, consistent with the graph's frozen-attention assumption; normalisation denominators are frozen only when the range spans the whole model, in which case the intervention measures a pure direct (linear) effect on the logits. The alternative, **iterative patching**, lets downstream features re-encode their perturbed inputs and take new values, propagating knock-on effects through the real MLPs at the price of compounding reconstruction errors. Effects are read out both as changes in the output distribution and as downstream feature activations re-encoded from the perturbed forward pass, reported as percentages of their clean baselines. Because feature dictionaries are imperfect (reconstruction error, incomplete feature recruitment), steering at $|M|$ beyond the observed activation is often required before behavioural effects appear; the paper typically suppresses source features to $M = -1$ or beyond and drives donor features at several times their donor-prompt activation.

Reimplementation. Our implementation realises the above for HuggingFace Qwen3 checkpoints with per-layer transcoders: a capture pass records attention patterns, RMSNorm denominators and MLP outputs; hooks then replace MLPs with transcoder reconstructions plus constant error nodes, substitute frozen-pattern attention (V/O only) and frozen-denominator norms; batched `autograd` computes Eq. (4) for hundreds of targets per backward pass. Pruning follows Section 2.3 exactly, and interventions follow circuit-tracer's constrained semantics. An eleven-check verification suite plus three steering-parity cases establish numerical equivalence with circuit-tracer: node sets, edge weights, influence rankings, pruned graphs, and constrained-steering deltas all agree to floating-point noise (Table 2, Appendix A).

3 Experiments and Results

3.1 Prompts, behaviour, supernodes, and attribution graphs

All experiments use Qwen3-4B (36 layers, bf16) with pretrained per-layer JumpReLU transcoders (163,840 features per layer) [17] on a single A100-80GB; notebooks seed all RNGs and pin cuDNN to deterministic kernels at startup, and re-executions reproduce every reported number exactly.

Prompts and behaviour. The paper's prompts are raw completions ending on a content-bearing open quote; we run them exactly, with one attention-sink token prepended:

```
<|endoftext|>The opposite of "small" is "
```
first generated token: `large`

Because Qwen3-4B is instruction-tuned, a chat-template adaptation runs alongside (thinking disabled; the answer is again the first generated token):

```
<|im_start|>user
What is the opposite of "small"? Reply with only the word,
nothing else.<|im_end|>
<|im_start|>assistant
<think>

</think>
```

first generated token: `large`

The antonym task is intact in both formats and all three languages (large / grand / 大); all ten prompts (three antonym recipients plus the English synonym and hot donors, per format) and their answers are tabulated in Appendix B.

placeholder — example reviewed features (labels, top output logits, top-activating dataset examples); screenshot to be provided

Figure 1: Examples of reviewed supernode members: each feature's top output logits and top-activating dataset examples — the evidence the semantic selection matches against.

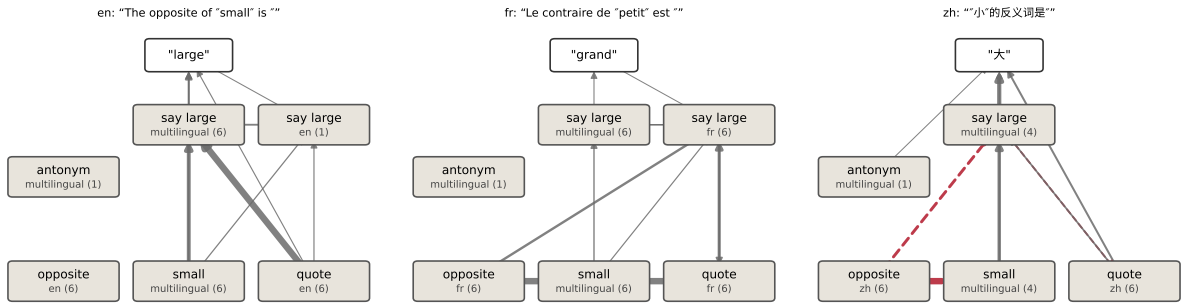

Figure 2: Simplified attribution graphs for the three raw antonym prompts. Boxes are the reviewed supernodes (member count shown); arrow widths are proportional to the aggregated signed edge weight between groups in the pruned graph (red dashed: net-negative). The paper's shared-core skeleton appears in all three languages.

Supernodes and attribution graphs. Supernodes are selected by *semantics, not influence rank*: a scan over the pruned graphs matches each feature's top output logits and top-activating input examples against per-concept lexicons, ranks candidates earliest-layer-first, and caps each group at six members; the chat and raw formats get fully separate selections, frozen in reviewed artifacts with per-graph activations and node positions per member, and enforced disjoint by (layer, feature). Figure 1 shows the evidence this selection matches against for example members. Figure 2 shows the resulting simplified graphs for the three raw antonym prompts: the paper's skeleton — language-specific *opposite* and *quote* input groups, a shared multilingual *small* group, and a shared *say large* stage feeding the answer — is present in all three languages, with 107 (chat) and 67 (raw) individual features appearing in all three languages' pruned graphs. One structural difference is visible immediately: the groups sit late (the raw *antonym* group has a single L34 member; *quote* members span L23–L34), where the paper describes early detection features. The chat counterpart of Figure 2 and full interactive-explorer views of all six antonym pages are in Appendix C.

3.2 Swap interventions

All swaps run the constrained-patching protocol of Section 2.4: source supernodes are steered to a negative multiple of their clean activation at their own node positions, donors are injected at an absolute multiple of their donor-graph activation, and the patch end layer ℓ is swept per job. Strength ladders extend tenfold beyond the paper's endpoint along the same source–donor coupling, with the paper endpoint marked in every panel.

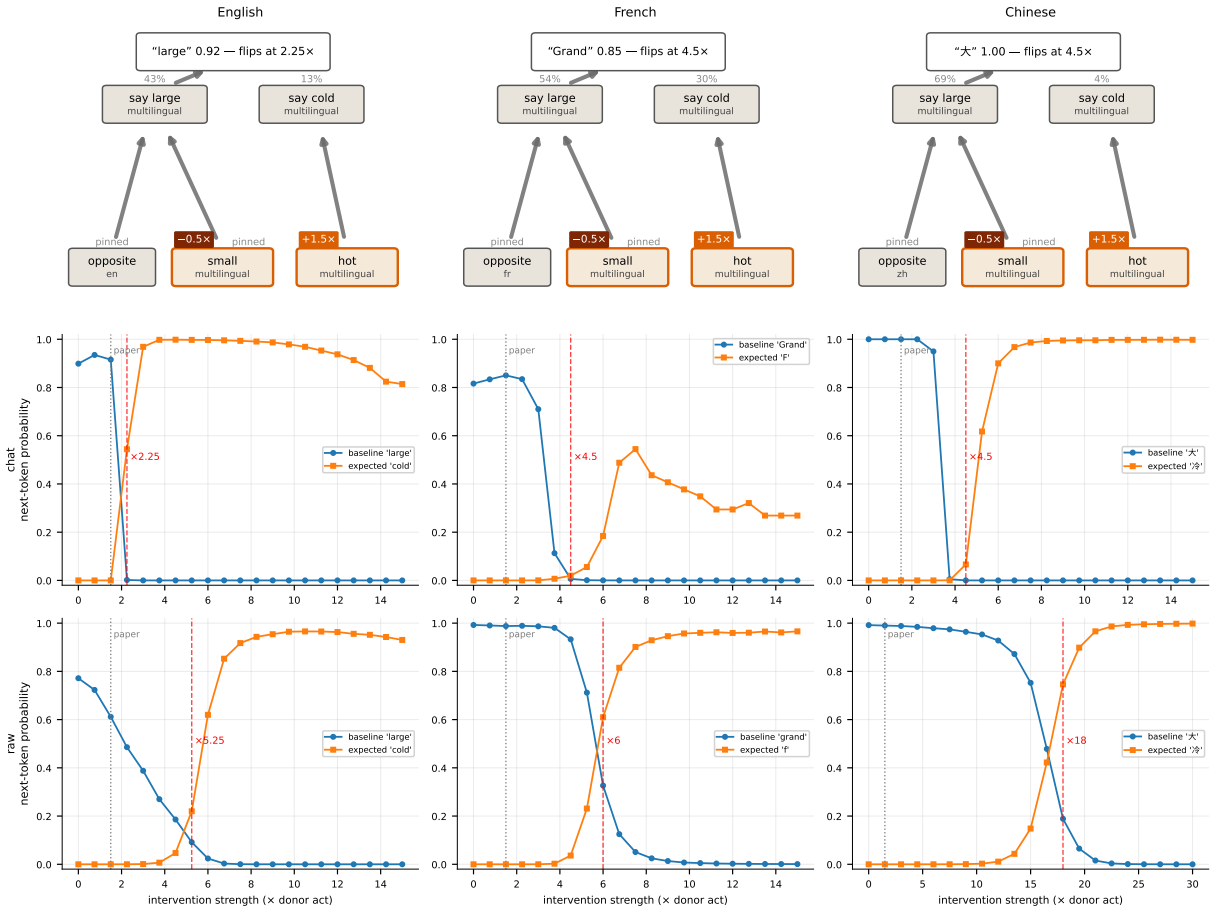

Figure 3: The operand swap. *Top:* intervention diagrams (chat) with measured supernode readouts at the paper endpoint; grey percentages are means of per-feature ratios, "pinned" marks groups at or below the swept ℓ. *Below:* strength ladders per format and language; red dashed lines mark crossovers, grey dotted the paper endpoint.

Operand swap. Steering small → hot reproduces in all six format–language configurations, at 1.5–12× the paper's strengths (Figure 3): nothing moves at the paper endpoint, crossovers sit at 2.25–4.5× (chat), 5.25–6× (raw en/fr), and 18× (raw zh), and where readouts are readable *say cold* recruits to 18–65% of its donor-prompt level while *say large* falls to 4–26%.

Language swap. Two of three directions reproduce at the paper's own scale (Figure 4): en→zh flips to 大 (0.636, crossover 5.25×), and fr→en turns the output English — *big* at 0.753, the paper's exact token, crossover 4.5× — while zh→fr never lands. Past ∼9× both working directions degrade into junk or wrong-language intrusions: this handle saturates. The quote features sit at L23–L34 (no early quote-position nodes survive pruning), so the causal handle is late rather than the paper's early detectors.

Operation swap. Steering antonym → synonym never crosses at any tested strength — up to ±29/30× — in either format (Figure 5), even though the chat side has full sweep room ($\ell \in [13, 35]$) and its readouts show the mechanism moving in the right direction but weakly (*say small* recruits to at most 37%; the output drifts to adjacent semantics such as *medium*). The paper itself flags the antonym→large interaction on Haiku as QK-mediated — "mediated by changing where attention heads attend" — and invisible to its frozen-attention method; since constrained patching freezes attention patterns by construction, we ran an extra experiment: the identical configuration in fully-propagating mode, once with patterns frozen and once with patterns free

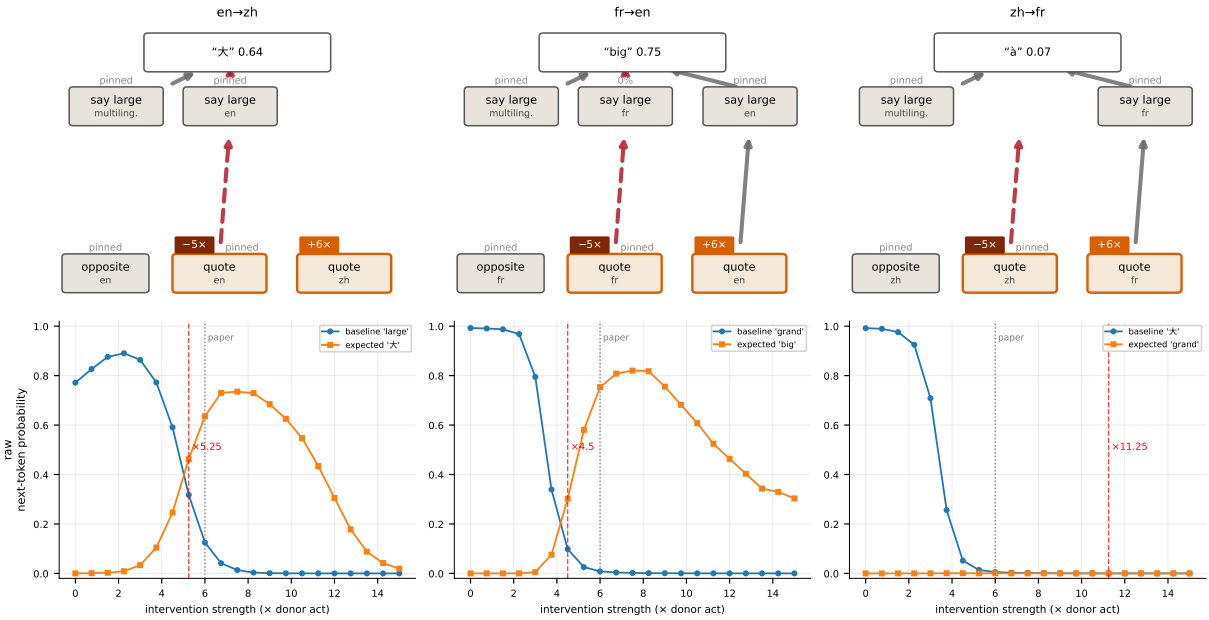

Figure 4: The language swap (raw prompts only; chat pages have no quote nodes). Conventions as in Figure 3; expected curves track the paper's outcome tokens (for en, *big* rather than the model's own answer *large*); the raw selection has no *say large (zh)* group, so that readout box is absent in the en→zh panel.

to re-form (Appendix D, Figure 10). Frozen stays flat; live lands the paper's exact tokens — *little* at 0.735 (en, crossover 6.75×) and *pet* at 0.515 (fr, 4.5×). The operation swap's channel is QK-mediated, so no frozen-pattern configuration can land it at any strength or selection. The three swaps thus separate cleanly by causal route: MLP-pathway (operand: works frozen, needs strength), direct-path (language: works at $\ell = 35$, saturates), and QK-mediated (operation: needs live attention).

3.3 Cross-lingual overlap, scale, and the default language

Cross-lingual overlap and scale. Following the paper, we compute the per-layer intersection-over-union of the feature sets active anywhere in the context for translated paragraph pairs, against an unrelated-pair baseline, on 18 register-diverse parallel paragraphs (authored in English and translated to French and Chinese — the paper's own corpus recipe). The paper's shape reproduces cleanly (Figure 6): overlap for translated pairs sits well above the unrelated-pair baseline, and the baseline-subtracted curves are low at both ends and peak mid-network, with mid-third means en-fr 0.198 > en-zh 0.175 > fr-zh 0.142 — the paper's pair ordering. The paper's *scale* claim does not reproduce on a same-recipe transcoder pair (4B vs. 8B, both 36 layers and 163,840 features/layer): mid-third changes are +6% / −3% / 0%, with no emphasis on the non-alphabet-sharing pairs the paper highlights.

The default language. The paper's Claude-specific claim is that English is the mechanistic default. Here the mean direct decoder effect of the shared say-large features on the three answer tokens is zh 0.766 > en 0.664 ≫ fr 0.309: *some* language is privileged, but for EN/ZH-trained Qwen3 the role is shared between English and Chinese.

3.4 Verdicts

Table 1 condenses the comparison. The paper's central structural claims transfer; its interventions transfer in proportion to how much their causal route depends on machinery the protocol holds

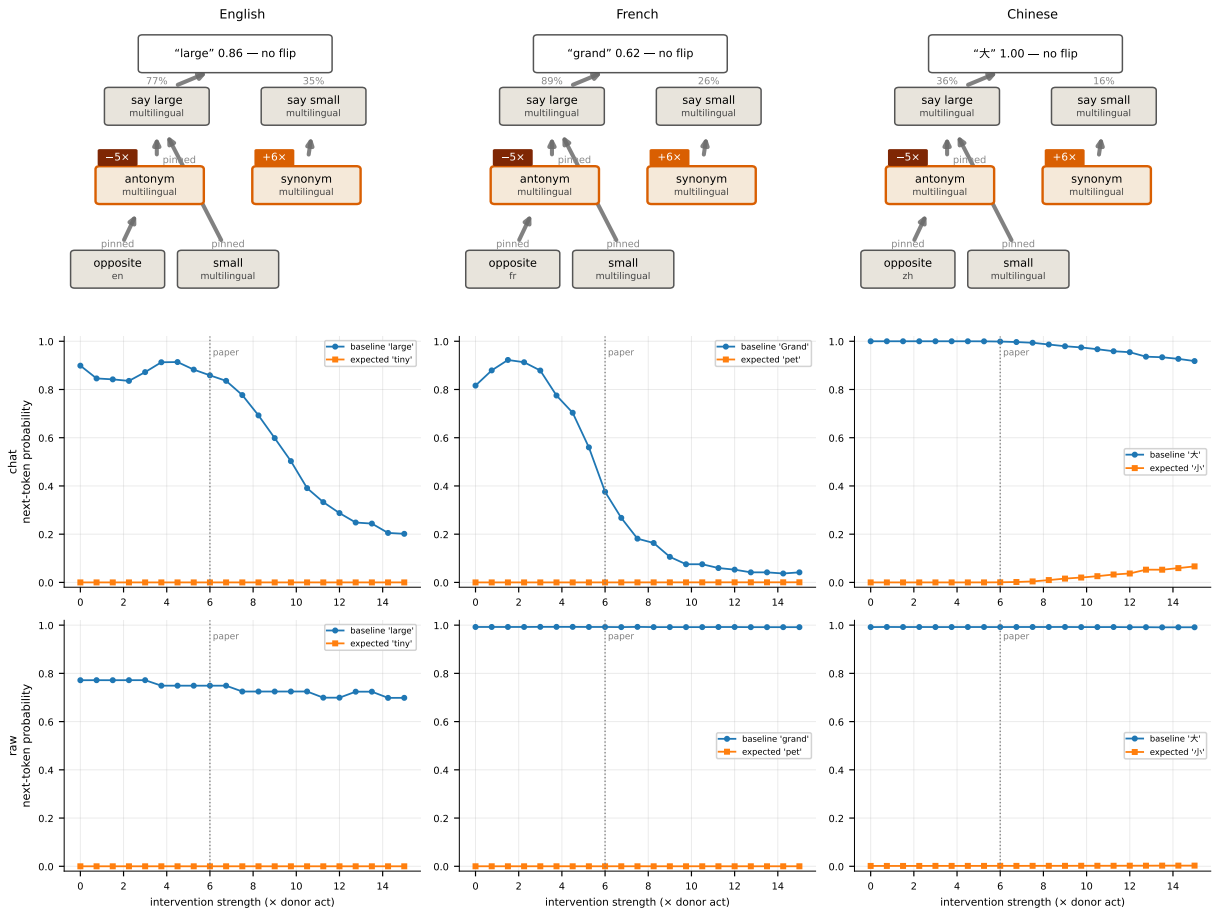

Figure 5: The operation swap under the constrained protocol: no crossover in any configuration. Conventions as in Figure 3; the frozen-vs-live attention probe is in Appendix D.

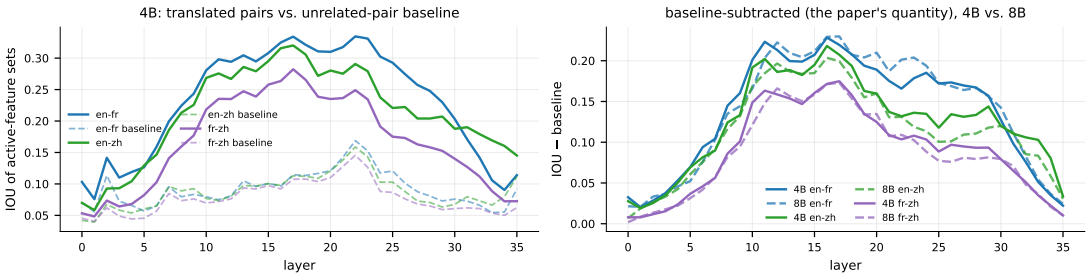

Figure 6: Cross-lingual feature overlap by layer. *Left:* 4B overlap for translated pairs (solid) against the unrelated-pair baseline (dashed); the residual baseline reflects the shared chat scaffold and the dictionary's granularity. *Right:* baseline-subtracted curves, 4B (solid) vs. 8B (dashed, same-recipe transcoders). The paper's inverted-U and pair ordering reproduce; its cross-scale growth does not.

fixed — fully for the MLP-pathway operand swap (given strength), partially for the direct-path language swap, and not at all for the QK-mediated operation swap.

paper claim (Claude 3.5 Haiku)	Qwen3-4B measurement	verdict
behaviour big / grand / 大	large / grand / 大, both formats	reproduced
shared multilingual core	107 (chat) / 67 (raw) features in all three pruned graphs	reproduced
operand swap at $-0.5\times/+1.5\times$	flips in all six configurations, crossovers 2.25–18×	reproduced*
language swap, three directions	2/3 at the paper's scale; saturates past $\sim 9\times$	partial
operation swap at $-5\times/+6\times$	no flip $\leq 30\times$ frozen; flips once patterns re-form	not reproduced†
overlap: mid-network peak, en-fr highest	$0.198 > 0.175 > 0.142$, ends ≤ 0.03	reproduced
English as mechanistic default	zh $0.766 >$ en $0.664 \gg$ fr 0.309	model difference
scale: 8B > 4B overlap	+6% / −3% / 0%	not reproduced

Table 1: Verdicts vs. Lindsey et al. [3]. *At 1.5–12× the paper's strengths. †Blocked by frozen attention patterns; the channel is QK-mediated (Section 3.2).

4 Discussion

[Placeholder — to be written; budgeted at ≈350 words.]

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A Verification against circuit-tracer

Two scripts certify the reimplementation against circuit-tracer on shared inputs (Qwen3-0.6B, fp32). The graph suite checks forward exactness of the local replacement model, an exact linearity identity for the frozen-linear backward, parity of salient logit selection, and a full cross-check against `circuit_tracer.attribute()`; the intervention script compares constrained-steering *deltas* across the two implementations for the three primitives the reproduction uses (a single negative steer, a two-layer coupled suppression stack, and an absolute-value donor injection). All eleven graph checks and all three steering cases pass (Table 2).

check	result
forward exactness (local replacement vs. model)	$\max \lvert\Delta\text{logits}\rvert = 1.2 \times 10^{-5}$
linearity identity of the frozen backward	max error 1.7×10^{-5}
200 emitted edge weights vs. plain `autograd`	$\max \lvert\Delta\rvert = 3.7 \times 10^{-5}$
salient-logit selection	identical tokens and probabilities
feature node sets	Jaccard 1.0
edge weights on 2.44M shared pairs	cosine = Pearson = 1.0000
influence ranking	Spearman 1.0000
pruned node sets (0.8/0.98)	identical (337 = 337)
steering deltas (3 intervention primitives)	$\max \lvert\Delta_{\text{ours}} - \Delta_{\text{CT}}\rvert = 6 \times 10^{-5}$, cosine > 0.99

Table 2: Numerical verification of the reimplementation against circuit-tracer.

B The ten prompts and their answers

graph	format	prompt (chat: the user line of the templated input; raw: sink token prepended)	answer
`antonym_en`	chat	`What is the opposite of "small"? Reply with only the word, nothing else.`	large
`antonym_fr`	chat	`Quel est le contraire de "petit" ? Réponds avec un seul mot, rien d'autre.`	Grand
`antonym_zh`	chat	"小"的反义词是什么？只回答一个词，不要其他内容。	大
`synonym_en`	chat	`Give one synonym of "small". Reply with only the word, nothing else.`	tiny
`hot_en`	chat	`What is the opposite of "hot"? Reply with only the word, nothing else.`	cold
`raw_antonym_en`	raw	`The opposite of "small" is "`	large
`raw_antonym_fr`	raw	`Le contraire de "petit" est "`	grand
`raw_antonym_zh`	raw	"小"的反义词是"	大
`raw_synonym_en`	raw	`A synonym of "small" is "`	tiny
`raw_hot_en`	raw	`The opposite of "hot" is "`	cold

Table 3: The ten graph prompts and the model's answer (first generated token). Not shown: the French and Chinese synonym behaviour used as operation-swap targets in Section 3.2 is degenerate — the model echoes the operand (petit → pet, 小 → 小) in both formats — and the hot-antonym targets are cold / F · f(roid) / 冷.

C Attribution graphs: chat format and explorer snapshots

Figure 7 is the chat-format counterpart of Figure 2 (chat pages have no quote-position nodes, so the *quote* groups exist only on the raw side). Full attribution graphs (all pruned nodes and edges, with the reviewed supernodes pre-loaded as groups) are best inspected in the interactive explorer shipped with the repository; the snapshots below show the six antonym pages.

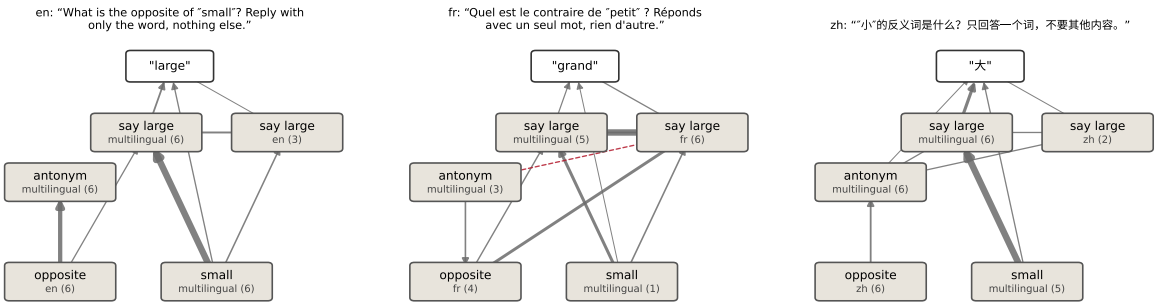

Figure 7: Simplified attribution graphs for the three chat-format antonym prompts (conventions as in Figure 2).

placeholder — explorer screenshots, chat antonym pages (en / fr / zh)

Figure 8: Explorer views of the chat-format antonym graphs with the reviewed supernodes as groups.

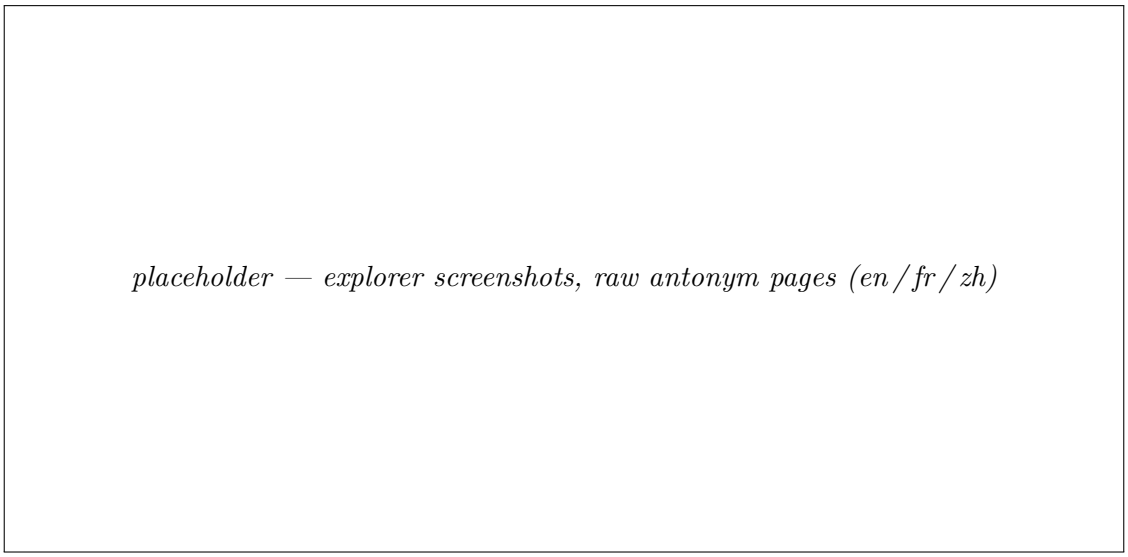

Figure 9: Explorer views of the raw open-quote antonym graphs (the paper's format).

D The operation swap with live attention patterns

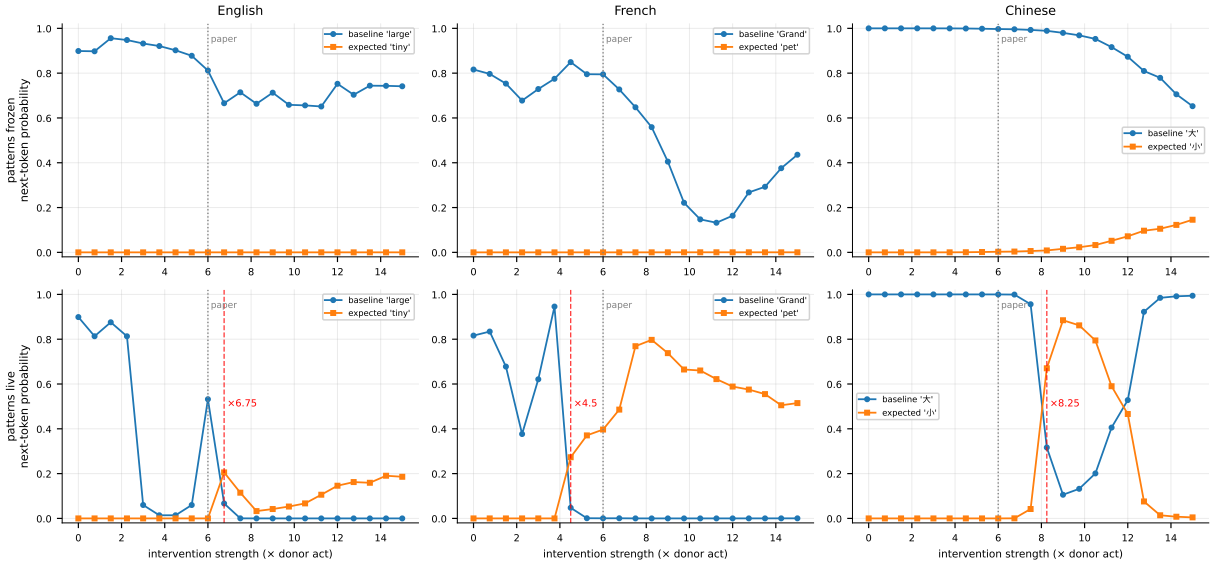

Figure 10: The frozen-vs-live attention probe for the operation swap (Section 3.2): the paper-faithful chat configuration in fully-propagating mode with attention patterns frozen (top) vs. free to re-form (bottom). With live patterns the paper's exact outcome tokens appear (*little*, *pet*); frozen, nothing moves — the swap's channel is QK-mediated. The en curves track the model's own chat answer "tiny"; the top token past the crossover is *little* (0.735 at 15×).